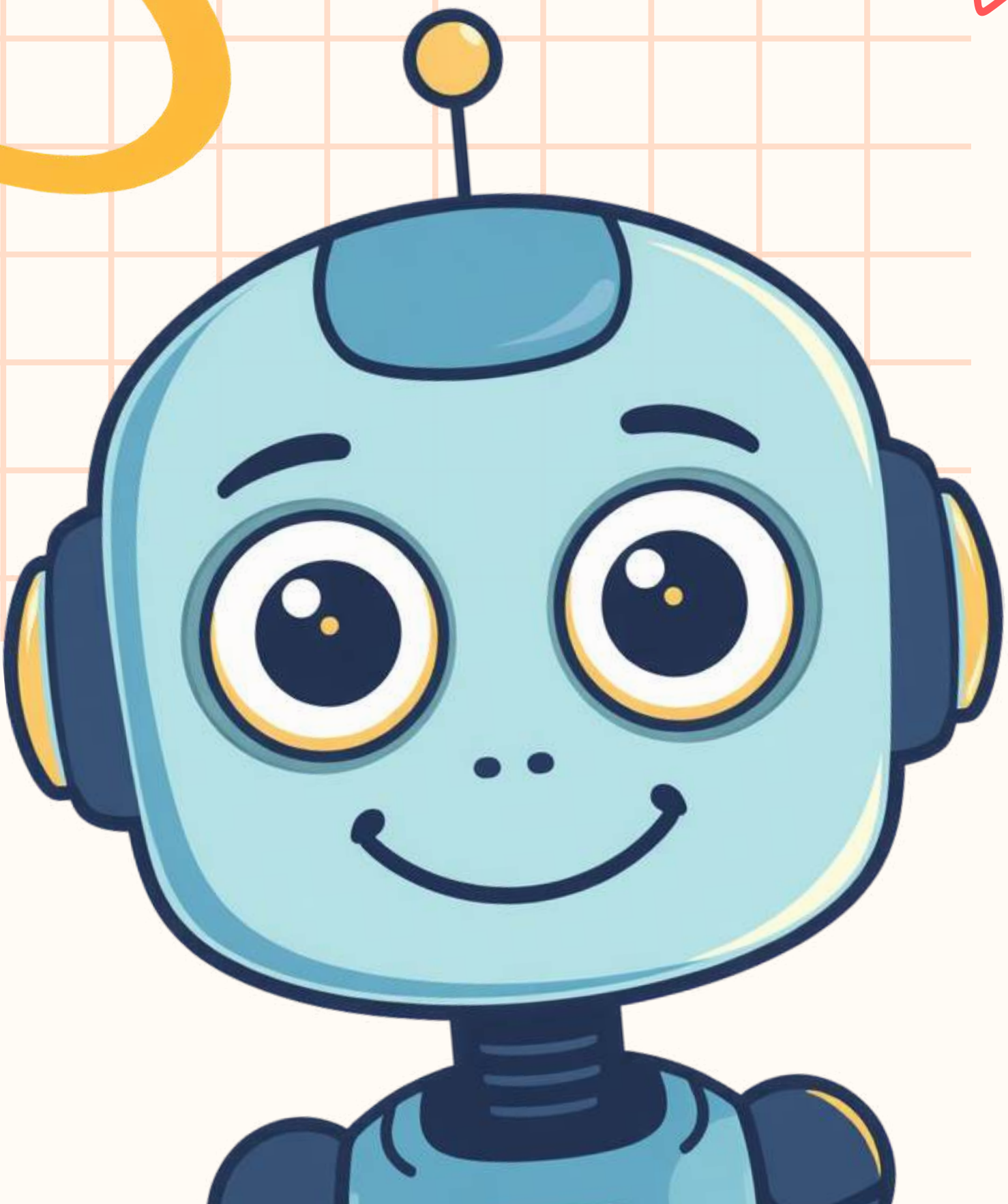


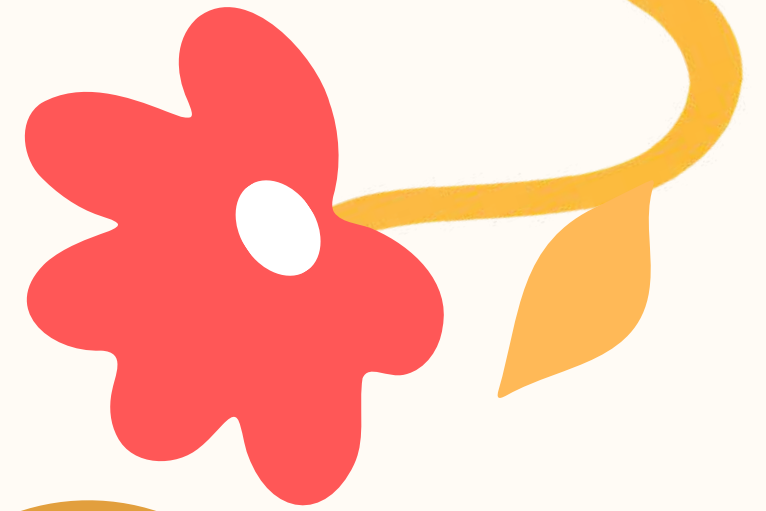


MEET THE ROBOT BRAIN!

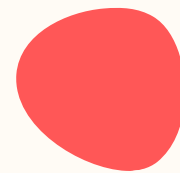


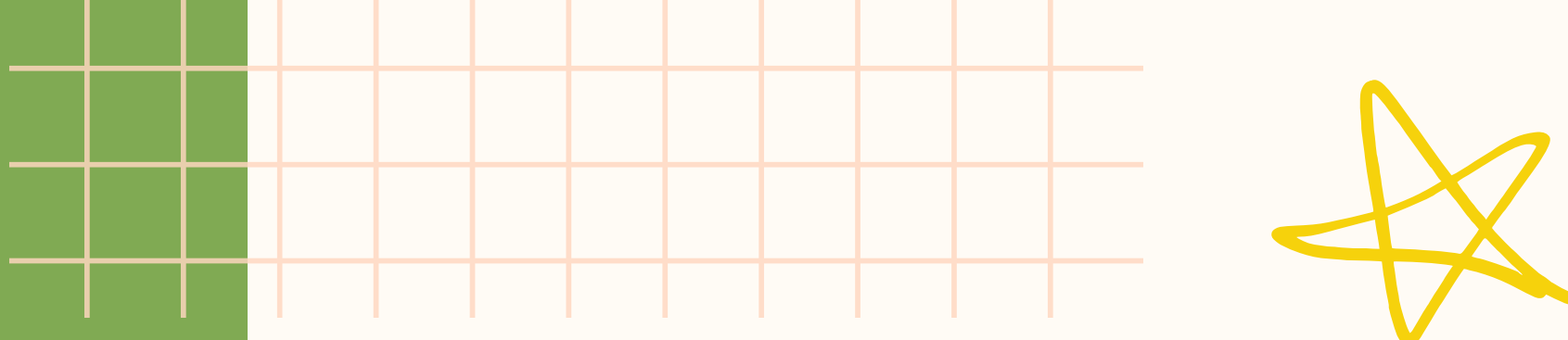


Let's Start with Tasty Indian Snacks!



Just like these snacks are different, machines learn patterns differently too! Let's explore how.



A decorative orange grid is in the top left corner, and a yellow star is in the top center.

What is Machine Learning?

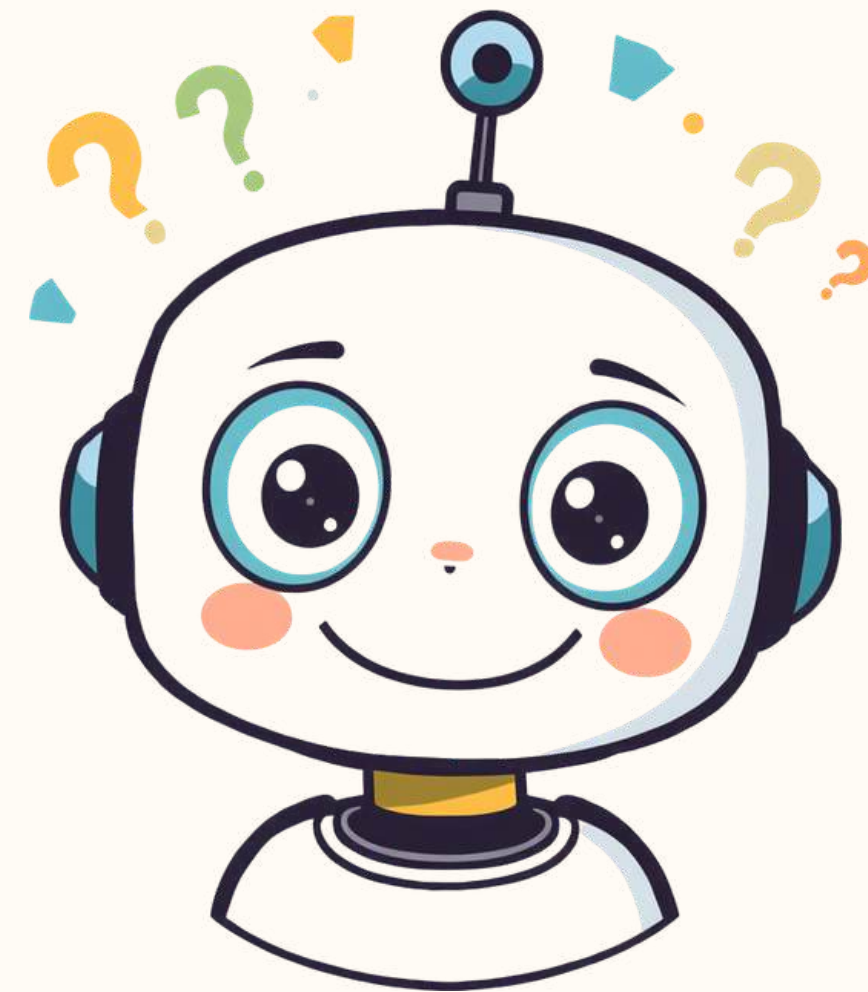
Teaching computers to learn from examples – not just follow fixed rules!

A small blue star is to the right of the text.A green wavy line is above the text.

Computers learn from data, just like we learn by practice.

A green wavy line is above the text.

They find patterns all by themselves – no strict rules needed!



Traditional Programming vs Machine Learning



Traditional Programming

- Write strict rules and instructions
- Example: IPL match rules are fixed!
- Computer only does what you tell it



Machine Learning

- Learn from examples and patterns
- Example: Sorting Alphonso mangoes by weight and colour
- Computer finds patterns by itself!



Features & Labels – What Does the Machine See?

Features are clues the machine uses.
Labels are the answers it learns to predict!

Features (X): Overs left, Runs needed, Wickets remaining.

Label (y): Match outcome – Win or Lose!



OVERS	RUNS	SCORE	BAERL
3	31	22	31
5	6	7	36
23	31	8.4	10





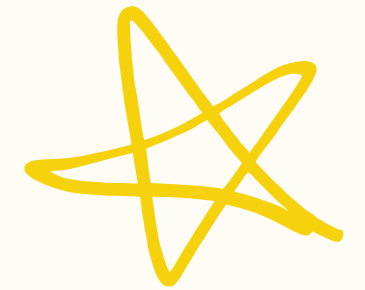
Supervised Learning – Learning with a Teacher!



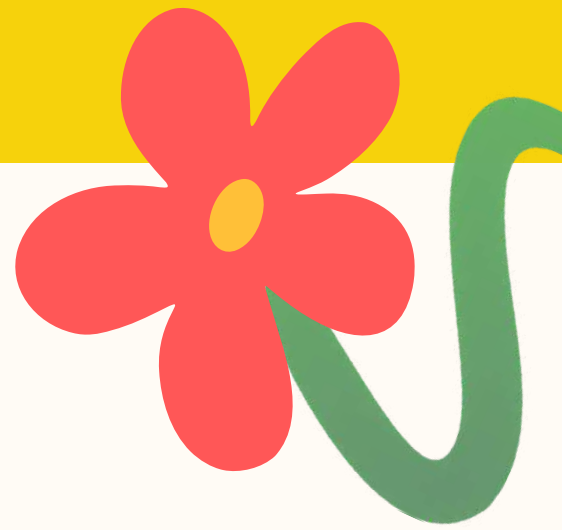
Supervised learning means the machine learns with a teacher showing it the correct answers.

Just like a cricket coach teaches a player which shots to play – the machine learns from labeled examples and gets better over time!

- ✓ Teacher gives correct answers
- ✓ Machine finds the pattern
- ✓ Machine predicts on new data!

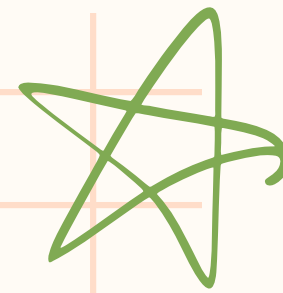


The Scikit-Learn 3-Step Recipe



1

Create & Train the model!
`model = DecisionTreeClassifier()`
`model.fit(X, y)`



2

Make predictions!
`predictions = model.predict(X_new)`
Easy, like a recipe!



Sorting Alphonso Mangoes!



The machine looks at two features: Colour Score (how yellow/orange the mango is) and Softness. It learns to classify each mango as Ripe or Raw – just like a fruit expert!

Features (X):

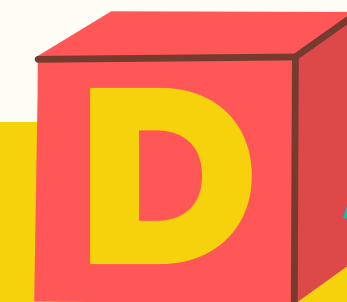
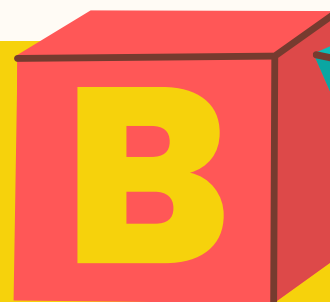
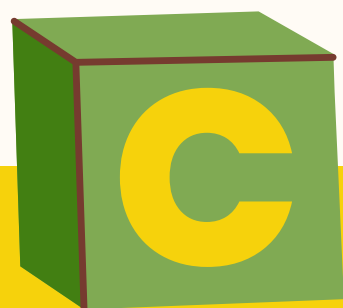
- Colour Score
- 👉 Softness Level

Label (y):

Ripe or Raw?



The machine finds the pattern and sorts mangoes all by itself – no rules needed!





Code-Along: Predicting Cricket Team Selection!



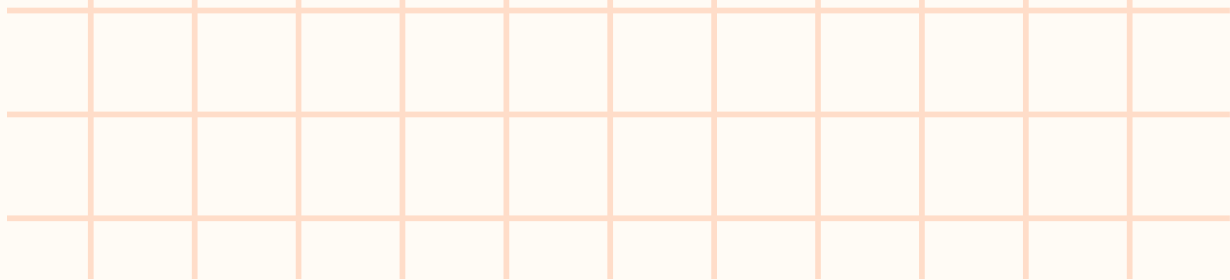
Dataset: Practice hours & batting average as features.
Target label: Selected or Not Selected.



```
X = [[5,30],[3,15],[8,45],[2,10]]  
y = ['Selected','Not Selected',  
     'Selected','Not Selected']  
model = DecisionTreeClassifier()  
model.fit(X, y)  
print(model.predict([[6, 35]]))
```

Let's see if our player gets selected! Type
the code and run it yourself!





Evaluating Predictions – Oops, Mistakes Happen!



Sometimes machines make mistakes if their data is weak or wrong!

- If we train only with right-handed players, the machine might be unfair.
 - Bad or missing data leads to wrong predictions.
 - More balanced data = smarter, fairer machine!
- 

Garbage In, Garbage Out!

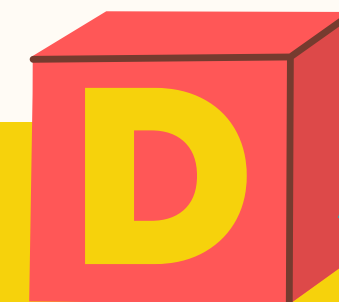
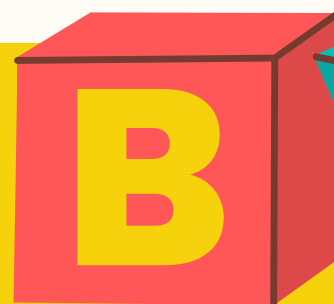
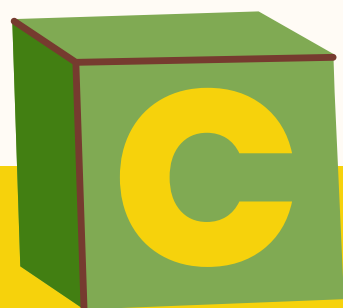


"If the data is garbage, the machine learns garbage." Biased data leads to unfair results. If we only train with data from one type of player, the machine won't be fair to everyone!

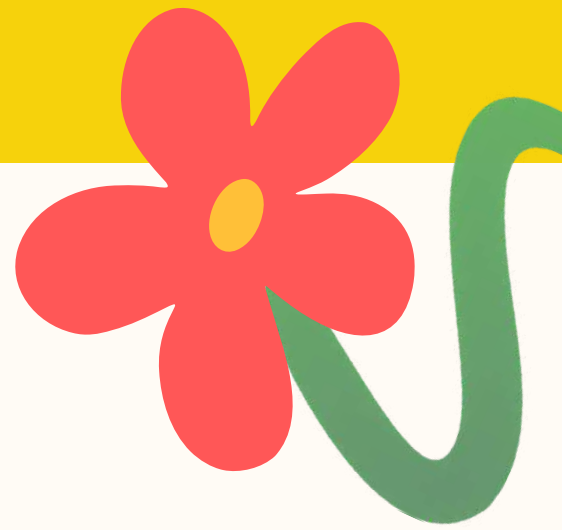
Bad data = bad decisions. Always check your data for mistakes and missing pieces!



Think about fairness! Would it be fair if the machine only learned from one kind of student?

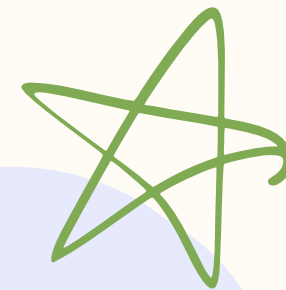


Recap & Quiz Time!



1

ML learns from examples, not rules.
Features are clues, labels are
answers. Supervised learning uses a
teacher!



2

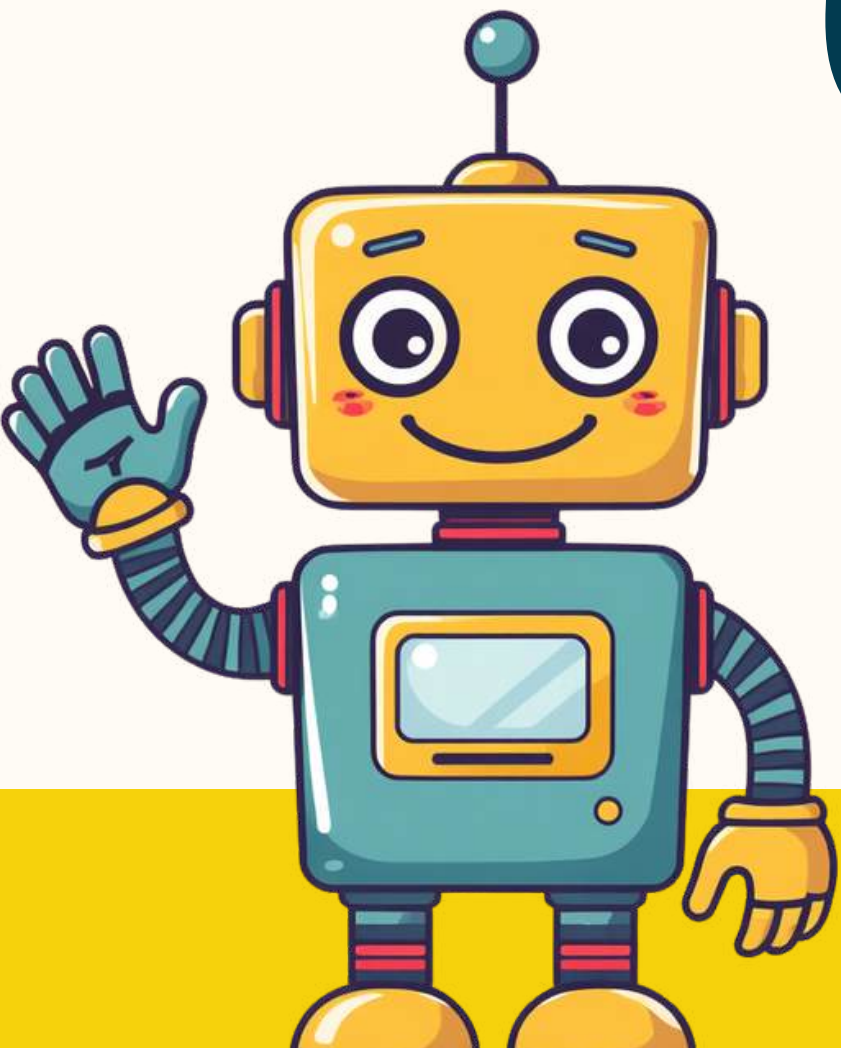
Quiz: What is a feature? What
does fit() do? Why is good data
important?



Your Take-Home Mission!

- Predict if it will rain in Mumbai for cricket practice tomorrow!
- Use the ML ideas you learned today.
- Keep exploring and have fun – you are a Machine Learning star!





Great Job Today!

